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Campaign Proposal – *Persuasive Technologies*
'Struddy': *Right Time, Right Help with Wearable Data*
A Just-In-Time Adaptive Intervention for Personalized Stress Detection and Early Support

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I. Introduction

The fact that people are experiencing high levels of stress in their daily lives, often without being fully aware of where it comes from or when it arises, is nothing new. In today's society, leading a stressful life has increasingly come to be viewed as the norm. The constant pressures, whether from work, social expectations or personal responsibilities have become so ingrained in our daily existence that they are typically no longer questioned. Instead, they are accepted as an unavoidable part of modern life, leaving little room for reflection on the consequences of this widespread, enduring stress (Jackson, 2014). However, we also know that prolonged periods of stress can lead to serious health problems, including cardiovascular disease, reduced sleep quality, anxiety, burnout, and dysregulation of the autonomic nervous system (McEwen & Akil, 2020; Vrijkotte et al., 2000). Chronic work-related stress, for example, affects autonomic regulatory processes (*heart rate variability [HRV], blood pressure, respiratory function, thermoregulation etc.*), ultimately undermining physical health.

Research shows that by only tackling people's peak stressful moments during the day (e.g. *moments of high arousal, measured via biometrics*) with short, mindful interventions (meditation, breathing exercises, muscle relaxation), we can improve their mood and lower their stress hormone production within a few days or weeks (Yadav et al., 2012; Beute & De Kort, 2018). This indicates that wellbeing can be improved by addressing these peak stressful moments throughout the day, in a relative accessible manner. Research supports this, stating that **feeling supported** lowers perceived stress levels significantly: "Even in professional environments, such as among clergy or in the workplace, the feeling of support leads to lower self-reported stress, even when accounting for other personal and environmental factors." (Ulrich Leicht-Deobald et al., 2018; Francis et al., 2018).

At the same time, currently over 454 million people use smartwatches worldwide, which is a 41% increase over 2024 (Kumar, 2025). These devices continuously collect physiological and behavioural data, including heart rate, heart rate variability (HRV), sleep, activity levels and more. This technology has demonstrated potential for early detection of health issues through real-time monitoring (Sailesh & Srilatha, 2019). Especially health-conscious users tend to use such devices (Chandrasekaran et al., 2020), placing that demographic in a unique position to detect stress/arousal patterns within their own bodies.

However, the full potential of this enormous amount of wearable data remains largely untapped. Most existing brands and apps (*such as Apple Health, Fitbit etc.*) focus on tracking individual parameters without offering any action-oriented feedback. They provide a nice overview of all the forementioned statistics and while these 'static' options offer valuable insights on their own, users are still left to identify patterns and draw conclusions themselves.

What is missing is a system that automatically integrates and analyses multiple measurements to detect changes in stress levels, autonomous from the user. By providing **real-time** insights through clear and personalized feedback, technology can evolve from merely tracking health and providing the data in an overview, to actively supporting users in managing their stress. It should be seen as a ‘buddy’ (*cfr. Struddy = stress + buddy*) that is constantly with you, providing positive feedback and suggestions to improve the user’s perceived stress levels. This way, wearables become an even more vital tool in preventive healthcare. Our proposal tries to change that by integrating real-time intervention options to act upon certain biodata, which will be discussed at length in this paper.

2. Technology

Chosen technology

A Just-In-Time Adaptive Intervention (JITAI) is a behavioural intervention that uses real-time data to deliver the right support at the right moment, based on a person’s changing context and needs. It adapts to internal states (*i.e. stress*) to provide timely help, often through mobile technologies, especially during moments of vulnerability or opportunity for behaviour change (Nahum-Shani et al., 2018). Struddy integrates this persuasive technology along with the diverse data points, transforming the separate numbers into clear, actionable insights that help users better understand and manage their stress. We offer a more holistic perspective. Wearables are excellent in detecting fluctuations in biometrics. We will leverage that to offer the user personalized messages, along with mindfulness-based intervention options to lower their arousal.

JITAI framework

JITAI contains several different components which will be discussed in this paragraph.

Proximal outcomes describe the more immediate desired goals of our intervention, such as lowering the stress levels at a given moment in time. These more short-term outcomes refer to the specific actions JITAI takes to decrease stress levels during the day, at these moments of arousal elevation. We expect that users who receive interventions will experience shorter stress peaks and less intense stressful moments compared to those who do not use the application.

The distal outcomes are the long-term goals our intervention wants to achieve. Stress is often a collection of different aspects and hard to pinpoint when you’re going through it. Our platform will alert the user at the right moment, so that each user gains timely insights into why and when they’re experiencing stress. It is about both **reducing stress** and **recognizing** the signs of stress in daily life. Ultimately, our goal is to improve the user’s overall subjective well-being by helping them manage and reflect on their stress experiences more effectively.

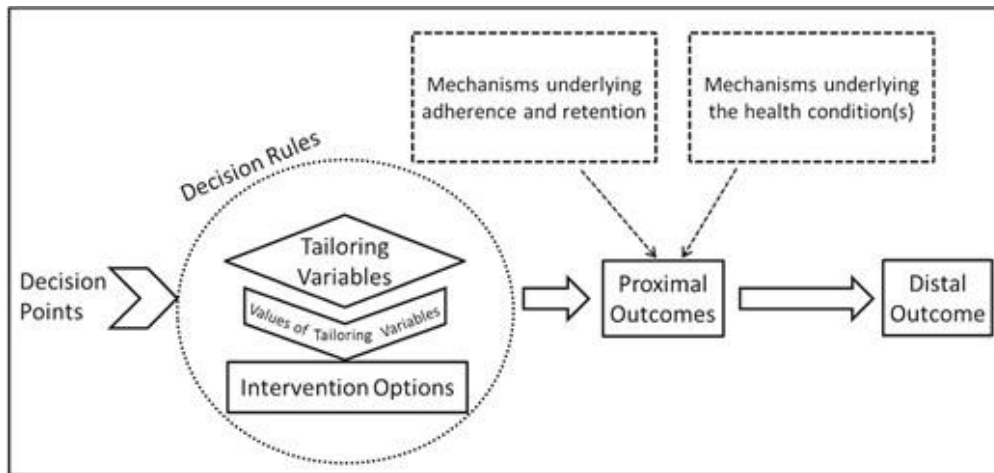


Figure 1 Conceptual model of JITAI components (Nahum-Shani et al., 2018).

Within JITAI, those stressful moments are described as **‘vulnerable states’**. These are the best moments to intervene, since the individual is more susceptible to negative health behaviours and most likely to respond to support, making the intervention timely and impactful (Nahum-Shani et al., 2018). These states dictate when and how the system will respond, based on the tailoring variables.

Those **tailoring variables** provide the information to measure when one finds themselves in a vulnerable state. It helps the system decide when to offer support. To know when a person is experiencing a stressful moment, the system needs information like HRV to be able to detect these states. “HRV is a measure of your autonomic nervous system that is widely considered one of the best objective metrics for physical fitness and determining your body’s readiness to perform.” (Whoop, 2024). It refers to the small changes in the time between each heartbeat. A healthy heart does not beat like a metronome. Instead, the intervals between beats constantly change. This natural variation allows the cardiovascular system to quickly adapt to sudden physical or mental challenges and maintain balance in the body. So, somewhat contradictory, the higher the variability, the better the adaptation to a flight or fight response (Shaffer & Ginsberg, 2017). A lower HRV is an indication for more arousal, linked to elevated stress levels. So the more variation, the more we are prepared for certain fluctuations in arousal. HRV can be measured "in the moment" using short-term recordings.

HRV has thus changed to an apparently simple point-of-care method obtained within 2–10 min. with potential clinical value for the patients regarding risk stratification, individual therapeutic strategies, and even therapeutically in the form of HRV biofeedback. (Ernst, 2017)

Metrics

Research has shown that certain HRV metrics, such as Root Mean Square of Successive Differences (rMSSD) and High Frequency (HF) power, can already be reliably assessed from ultra-short recordings of just 10 to 30 seconds. Research shows that these short-term measurements are reliable to measure stress (Castaldo, 2019). This makes HRV particularly suitable for real-time applications or stress monitoring during brief measurement windows. rMSSD is a measure that looks at short-term variations between successive heartbeats, giving insight into how flexible the heart rate is. HF power reflects faster changes in heart rate and provides information about the influence of the nervous system, especially the part responsible for rest and recovery (Ernst, 2017). rMSSD values of 25 ± 2 ms and 35 ± 2 ms are cited as possible thresholds below which the risk of psychological stress and depression is increased in adults. Individuals with an rMSSD lower than these values have a significantly higher likelihood of stress-related complaints and depression (Watanabe, 2024), so we will use this threshold for our system. While an elevated heart rate or variated HRV never 100% correlates to stress, research shows that these biometrics are the most reliable to detect arousal for our purposes (Chia et al., 2024). Long story short: a low HRV indicates higher stress levels and poorer mental health (Chia et al., 2024) and is generally described as the best and most reliable measurement for stress in wearable devices (Can et al., 2020).

The tailoring variable (HRV) will be measured via an adequate smartwatch (*Apple or Android*). Not every smartwatch tracks each measure as accurately, Apple Watch and Fitbit devices only show a small error percentage in heart rate measurements, with errors staying between 1 to 9% (Wallen et al., 2016). Our system offers an app that will be available on iOS and Android wearables, covering most of the market shares in the wearable space. Knowing our metrics are sufficiently reliable, we can shift focus to the interventions. After recognising the vulnerable states based on the tailoring variable, the system can generate personalised interventions to lower the stress level at the moment itself.

Struddy's interventions with JITAI

Research shows a variety of different intervention options with varying efficacies. Mindfulness-based stress reduction (MBSR) shows great results in lowering stress levels (Nykliček et al., 2013). MBSR interventions include exercises like **body scans**, where users focus on the sensations in their body to shift focus from stressful thoughts to the own body, **walking meditation** where paying attention to sensory cues (sounds, sensations, posture and movement) shift the focus from the stressors to the present body and **breathing exercises**, where the user takes a few minutes to focus on their breath, which is shown to lower stress levels significantly (Birdee et al., 2023). The effects of these interventions can be measured by looking if the heart rate, and subsequently the HRV change after performing the exercises. Lastly, based on research, we suggest the option to **play relaxing sounds**. The user will be able to choose between a variety of different sounds (forest sounds, white noise, flowing oceans, ...) to calm the mind.

Relaxing music and natural sounds are generally more effective at reducing stress than silence or other types of music, such as rap. For example, relaxing music and autonomous sensory meridian response (ASMR) sounds were found to calm subjects faster than silence (Fan & Baharum, 2024; Paszkiel et al., 2020).

We will select the forementioned four intervention options to give users variety in their treatment styles. Research also shows that mindfulness improves the capabilities to recognise stress patterns, which adds to the goal of making users recognising these stressful moments within themselves better and more clearly (Lindsay et al., 2024). All of these factors combined make the four options above the most scientifically sound options, while also offering enough variety for the user.

If Struddy detects a stressful moment, a supportive message is sent out. This is followed by a suggested intervention (one of four), that can be started immediately after getting the notification by selecting 'Start the exercise'. However, it is not always viable to carry out the intervention immediately (*e.g. when the user is in the middle of a meeting*). That's why there will be an option to select 'Remind me later', after which Struddy will check in again later to still suggest an intervention option, after the meeting has passed (see Figure 2).

In the settings of the app, users can select which type of messages they want to receive. They can pick between three categories: grounding-/calming reminders, validation and emotional support and light, soothing encouragement. These supportive messages can also vary. There will be a pool of different possible messages, so that users won't get the same notification every time. On the next page are examples of these supportive messages.

Possible supportive message interventions for Struddy
Grounding, calming reminders
It's okay to pause. Just take a deep breath, no pressure, just presence.
You don't have to fix everything now. Let the moment pass.
Your body is doing its best. One calm breath at a time.
Let your shoulders drop. Unclench your jaw. That's already enough.
Validation & emotional support
It's okay to feel off. You're allowed to have moments that feel heavy.
You're doing better than you think. Be kind to yourself.
You've made it through hard days before. You can handle this too.
Light, soothing encouragement
You're not alone. I'm here to remind you to take care of yourself.
Everything doesn't need to be perfect. You're allowed to just exist.
You are not your stress. It will pass.

Table 1 Possible message notifications from Struddy sorted by category/preference.

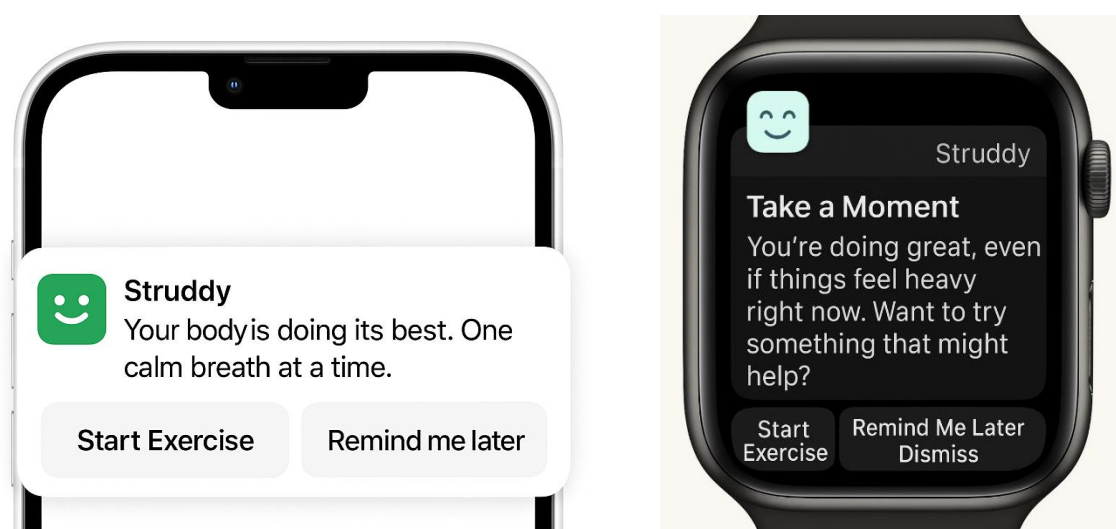


Figure 2 Mock-up notifications from Struddy

Lastly, at the end of each day, an automated message can be sent to users with a prompt such as “It seemed like a busy day, how are you holding up?”, along with another option to do an intervention exercise. This reflective question encourages users to pause and assess their emotional state. Importantly, even on less stressful days, sending a message like “Today seemed like a mindful day. Great job!” can be equally valuable to gain insights into (the lack of) stressors throughout the day. Acknowledging both high- and low-stress moments supports a more comprehensive understanding of individual stress patterns. This approach provides deeper insights into well-being, highlighting not just when stress is present, but also when it is notably absent. This will further integrate Struddy as an actual ‘buddy’ in the user’s journey in their stress management and recognition.

More on mindfulness for stress relief

Smartphone apps aimed at mindfulness have been shown to be effective in reducing stress, negative affect and irritability. A randomized controlled trial demonstrated that even brief use of a mindfulness app that offers similar interventions can lead to significant improvements, suggesting that digital platforms can play a valuable role in mindfulness training and stress reduction (Economides et al., 2018).

However, the use of such apps also presents challenges, particularly in workplace settings where app notifications can actually increase stress. Constant interruptions from notifications can elevate mental workload and hinder concentration, potentially leading to higher stress levels over time (Tams et al., 2015). Therefore, it is crucial to thoughtfully integrate mindfulness apps into daily routines, paying attention to how notifications are perceived and managed. By carefully tailoring notifications and being mindful of interruptions, as well as offering the option to postpone the notification, the positive impact of mindfulness apps can be maximized without causing unintended negative effects.

The benefit of JITAI over other technologies for this application

Compared to other technologies like AR, VR or chatbots, JITAI has a clear benefit for our specific goals. While other technologies assume that the user must decide when to use them, JITAI does ‘the guesswork’ for them. Like stated earlier, a JITAI solution works completely autonomously by gaining information from the sensors in the devices. No initial user input is necessary aside from wearing the watch, which the user is doing each day anyway. This takes away the need of the user to recognise that there is a stress-response, adding to the ease of use and unobtrusive nature of this technology. For other technologies, the user needs to already acknowledge that they are experiencing stress and then put on a headset, take out an AR-device or open a chatbot application. Our goal is to tackle the stress responses at the exact moment they arrive, without the need for the user to recognise them.

Furthermore, people often do not know exactly when they are experiencing stress. Their subjective sense of stress often does not align with the biological signals of stress in the body (Lupien et al., 2022). Stress is primarily experienced and recognised on a psychological level, and the degree of awareness about it varies from person to person. People can experience stress without being aware of their physical responses, or they may exhibit physiological stress reactions without feeling subjectively stressed (Lupien et al., 2022; Kemeny, 2003). That's why Struddy will help the user recognise their stress patterns faster, by providing real-time information about triggering stressful moments.

Research shows that recognising and understanding our own stress is the first crucial step toward implementing more effective coping strategies, developing problem-solving skills to address the sources of stress, and ultimately reducing psychological symptoms. In contrast, denial and avoidant behaviour tend to produce the opposite effect (Thoits, 2006; Gilhooly et al., 2016). Struddy will help users in this learning process – often indirectly – by letting them pause at stressful moments (by receiving a notification), adding to this recognition every time their stress/arousal is elevated. Over time, this repeated association through notifications helps them build a stronger internal awareness of their stress patterns (Wei Lü et al., 2016; Schiweck et al., 2022). In short: learning about what stresses us when, helps develop better skills to tackle stress in the future, strengthening our resilience over time. Struddy will help the user in this progress through JITAI.

3. Evaluation

This final section evaluates the effectiveness of Struddy through both main and process effects, incorporating quantitative and qualitative methods.

Experimental research

To evaluate the effectiveness of Struddy as a Just-In-Time Adaptive Intervention, an experimental research design will be set up. In this research, we will test whether the intervention actually leads to a change in our proposed main effects (the primary goals we want to achieve) and process effects (the mechanisms that explain why the intervention works).

Main effects: **stress reduction** and **stress recognition**

Struddy's interventions are tailored based on two main desired results. Firstly, we want to lower the user's stress/arousal level at the moment it elevates significantly, the so-called 'peaks'. This is operationalised as **stress reduction**. Secondly, Struddy aims to offer the user a better insight into when and how their stress levels rise. We want to give them better insights into their stress patterns, what triggers them and which signals are being set off during these stressful moments. This is operationalised as **stress recognition**. These are Struddy's premised main results.

The degree of **stress reduction** will be evaluated by both subjective and objective, physiological measurements. On one side, we will use the previously validated and trusted Perceived Stress Scale (PSS-10) to measure the perceived stress levels of users before and after the intervention. This scale has been tested and reviewed multiple times, and has an average Cronbach's Alpha between 0.74 and 0.91 (Lee, 2012). On the other side, we will analyse the physiological stress responses using numerical data from the wearable. This is stored within the app and, with permission, can be extracted for further analysis. This data includes heart rate and HRV, indicating arousal and stress and thus giving us a quantified idea of the stress levels of the user. Based on this data, we can track whether the users of Struddy experience less time under stress (higher HRV) compared to non-users, i.e. the control group.

Stress recognition is described as the user's ability to identify when and why they experience stress. As there is currently no validated questionnaire that specifically and exclusively measures the level of stress recognition in individuals, we chose to develop a self-constructed set of questions. Existing questionnaires around stress typically measure the level of stress (such as the PSS-10), but do not address the metacognitive awareness needed to recognise patterns in stress. These questions can be found below.

Stress recognition questions	1 (strongly disagree) - 5 (strongly agree)
1. I tend not to notice feelings of physical tension or discomfort until they really grab my attention.	
2. It seems I am “running on automatic,” without much awareness of what is bringing me worries.	
3. I can usually pinpoint what situations or behaviours in my day triggered my stress.	
4. I am aware of how certain habits or routines affect my stress levels throughout the day.	
5. I notice early warning signs in my body when I start to feel overwhelmed.	

Table 2 Self-constructed ‘stress recognition’ scale.

Process effect: **Compliance** and **feeling supported**

Besides the main effect, we also investigate the process effect. These consist of two components: **compliance** and **feeling supported**. These effects provide insight into the user experience of Struddy and can explain why the intervention works for one user and not for another.

Compliance refers to the extent to which users use the app as intended. Since Struddy uses an adaptive method that provides real-time notifications and exercises, it is crucial that users also respond to it in a timely manner, otherwise it misses its effect. We measure compliance using the data collected in the app, with permission of the participating subjects. The evaluation also looks at the percentage of notifications users take action on, the speed with which they respond to interventions, and the number of interventions that are actually completed in full. We also analyse whether users wear their wearable consistently; lack of biometric data may indicate low engagement. This data provides a clear quantitative picture of the extent to which users are actually using Struddy as intended.

Compliance can naturally only be measured post-intervention, and only in the experimental group. We can’t measure compliance before there was anything to comply to, or on a group that didn’t have to comply to anything. So this will only be measured post-treatment, and only for the experimental group.

Feeling supported measures the extent to which users feel supported by the intervention at moments of stress. This subjective feeling of support is important since it enhances the effectiveness of behaviour change and emotional wellbeing (van Brakel et al., 2023). When users perceive that the app is guiding them at the right time, it might strengthen their confidence in the technology and increases the likelihood that they will be open to the exercises offered. To quantitatively measure this sense of support, we rely on an existing scale that measures similar sentiments. The Cronbach's Alpha for this original scale was 0.93 (Schmidmaier et al., 2024). We selected the most fitting items for our application and chose to use the same 5-point Likert scale from before, to ensure consistency. The result of this selection can be found below in Table 3.

Feeling supported questions	1 (strongly disagree) - 5 (strongly agree)
1. The system showed interest in me.	
2. I trusted the system.	
3. The system considered my mental state.	
4. The system supported me in coping with an emotional situation.	
5. The system gave solutions.	

Table 3 Self-constructed 'feeling supported' question set.

These questions will be presented to the post-treatment experimental group, to measure how supported they feel after using Struddy. The control group won't receive this question set, since they didn't get to use the system (Struddy), so can't evaluate this variable.

Research design

Research will be carried out in two conditions: an experimental group that will use Struddy and a control group that won't. However, the control group will also have to wear a wearable to register the same biometrics as the experimental group to compare the data later on. Both groups will fill out the same set of surveys for the main effects, with both forementioned scales for stress reduction and recognition (main effects). They will both fill in the survey before the intervention (T1) and after (T2). Only the experimental group will go through the JITAI-intervention via Struddy. They will have to agree to using the app for a set amount of time, each day. The process effects will only be measured in the experimental group, as mentioned above.

Group	Pretest (T1)	Intervention	Posttest (T2)
Experimental condition	Stress reduction: PSS-10 scale+ numerical data Stress recognition: scale	Struddy (JITAI)	Stress reduction: PSS-10 + numerical data Stress recognition: scale Compliance: data from app Feeling supported: scale
Control group	Stress reduction: PSS-10 + numerical data Stress recognition: scale	—	Stress reduction: PSS-10 + numerical data Stress recognition: scale

Table 4 Overview of experimental research design

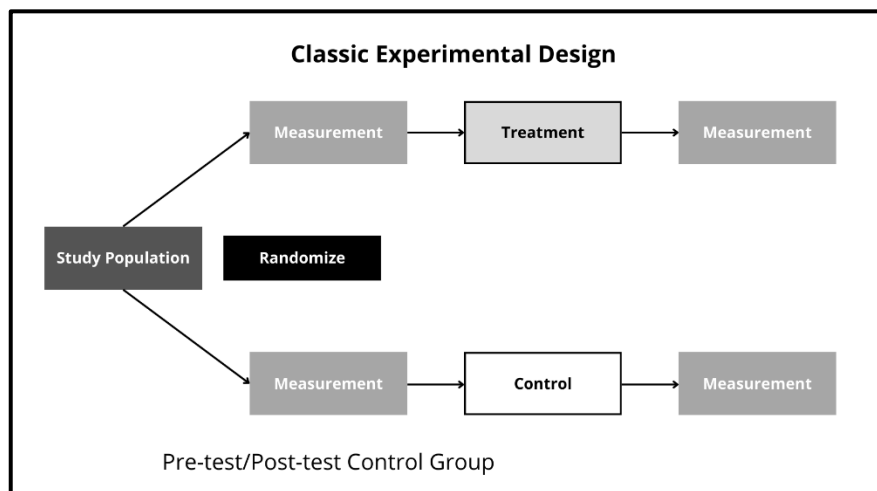


Figure 3 Schematic of experimental design.

To analyse the main effect, we will compare the changes in both scores (stress recognition and stress reduction) for both groups (pre- and post-test results), after which the experimental group will be compared to the control group. This allows us to gain insight into the causality of Struddy's effect on stress reduction and recognition.

More on the experimental design

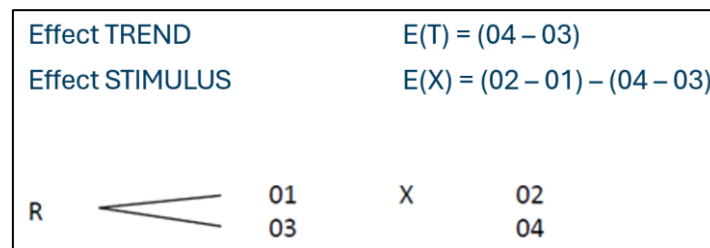


Figure 4 Experimental design

Both groups are tested at two different times: once before the intervention (X) and once after the intervention. In Figure 4, 01 and 03 are the measurement moments before the intervention, and 02 and 04 are the measurement moments after the intervention. This way, we can measure both the stimulus effect of the intervention (Struddy) and the trend effect of any external factors separately. The trend effect refers to natural changes that occur over time, independent of the intervention. For example, stress can decrease due to a variety of contextual factors such as reduced workload, holidays, ... that are independent from the use of Struddy. To measure this, we look at the change in the control group between the two measurement times. This translates into the following formula: **$E(\text{TREND}) = 04 - 03$** .

The stimulus effect is the actual effect of the intervention Struddy. We calculate the difference in the experimental group between 02 and 01 and subtract the previously calculated trend effect of the control group. The formula for the stimulus effect reads **$E(X) = (02 - 01) - (04 - 03)$** .

To decide the length of our experimental trail, we looked at a meta-analysis from Heber et al. (2017) on the effectiveness of computer-based interventions for stress reduction in adults. They examined the influence of the length of the testing phase on experimental outcome measures. Medium-long phases (5-8 weeks) were found to be most effective, with an effect size of Cohen's $d = 0.59$. Short phases (<4 weeks) were less effective, and long phases (≥ 9 weeks) produced no significant additional benefit. Based on these findings, we chose to run an experimental intervention with a duration of 8 weeks.

For our experiment, working adults with full-time day jobs (9am - 17pm) were chosen as the population to limit the variation in daily structure between participants. This target group tends to have somewhat predictable schedules, which means that stress moments occur slightly more consistently and the effectiveness of Struddy can be evaluated more reliably. Thus, external factors such as irregular working hours or sleep patterns might be better controlled.

To determine the required sample size, a power analysis was performed using the G*Power application. Taking into account an expected mean effect size (Cohen's $d = 0.5$), a significance level of $\alpha = 0.05$ and a statistical power of 0.80, a sample size of at least 68 participants (34 per group) was found to be required. Since possible attrition or incomplete data (estimated at ~10–15%) must always be taken into account, a total sample of 76 to 80 participants is aimed for. These will be randomly assigned to the control or experimental group.

An overview of our evaluation plan

Effect Type	What is measured	How it's measured
<i>Main effects</i>	Stress reduction	PSS-10 (pre- and post-test) Quantitative wearable data
	Stress recognition	Custom 5-item scale (pre- and post-test)
<i>Process effects</i>	Compliance <i>(engagement with app)</i>	App data: response rate, prompt interaction
	Feeling supported by the intervention	5-item Likert scale (post-test only)

Table 5 Overview of our evaluation plan.

Limitations and Future Considerations

Struddy faces several limitations that are important to acknowledge. Not all forms of stress are easily picked up through HRV alone. Low-arousal stress such as mental fatigue, emotional suppression or overthinking often go undetected, even though they can be just as impactful. Conversely, physical activity can trigger similar biometric responses, leading to false positives where Struddy interprets fatigue as stress.

User engagement is another challenge. While some users respond well to prompts, others may experience them as intrusive or even stress-inducing. The system also relies on a degree of intrinsic motivation: users must be willing to interact with the exercises and feedback and be willing to lower their stress in the first place. In addition, giving users the freedom to use any supported wearable introduces variability in data quality, with budget models typically being less accurate than high-end devices. Notification fatigue also remains a risk if Struddy becomes too present.

To address these challenges, we plan to integrate more contextual data, such as calendar events or location, to better assess when and how to intervene. This should help reduce both false positives and false negatives. We also intend to explore gamification strategies, like streaks, badges and goal setting, to increase motivation and retention. Finally, insights from the experimental study and user feedback will guide future updates, allowing Struddy to become more adaptive, accurate and engaging over time.



Struddy

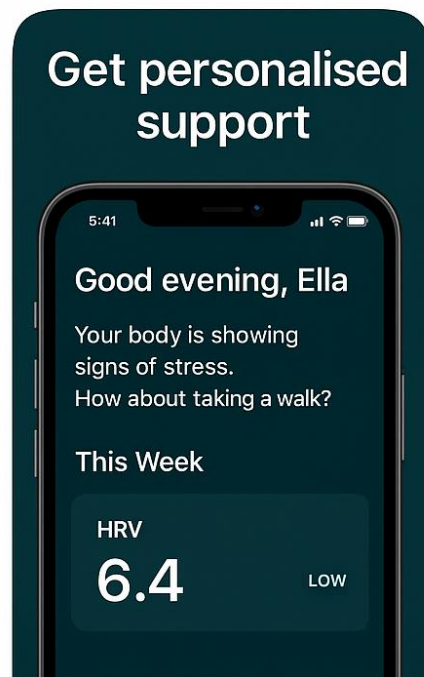
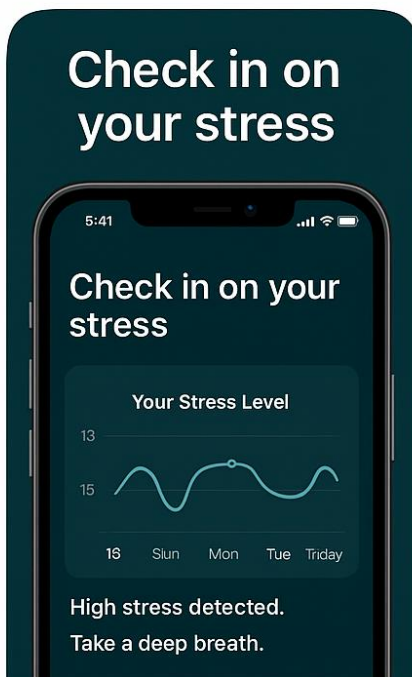
Understand and manage
your stress

Gain insights into your stress levels and
get personalized, just-in-time support to
help you feel better.

4,7
★★★★★

Health & Fitness
Health &

12+
Age



GET

Figure 5 Mock-up of Struddy on the App Store landing page.

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